Broken Web App Challenge

Pre-fix check commands (read-only) | Prepared by: Sukhada Wagale | 17 October 2026

Run these in AWS CloudShell or any terminal with the AWS CLI, in **us-east-1**. They only read information; nothing is changed. Replace values in <angle brackets>.

0. Setup: find names and IDs

```
export AWS_DEFAULT_REGION=us-east-1

# ALB URL from stack outputs (replace sukhada with your stack name)
aws cloudformation describe-stacks --stack-name sukhada \
  --query "Stacks[0].Outputs" --output table

# All resources the stack created, with their IDs
aws cloudformation describe-stack-resources --stack-name sukhada \
  --query "StackResources[].[ResourceType,LogicalResourceId,PhysicalResourceId]" --output table
```

1. Symptom check: what is failing?

```
curl -I http://<ALB_DNS_NAME> --max-time 10
```

Result	Usually means
Timeout	Issue 1 (ALB in private subnets)
503	Issue 4 or Issue 2 (no healthy / no registered targets)
504	Issue 3 (websg blocks port 80)

2. Issue 1: Is the ALB in public or private subnets?

```
# Which subnets is the ALB in?
aws elbv2 describe-load-balancers \
  --query "LoadBalancers[].[LoadBalancerName,Scheme,VpcId,AvailabilityZones[].SubnetId]"

# CIDR and name of every subnet (identify by CIDR, not the Name tag)
aws ec2 describe-subnets --filters Name=vpc-id,Values=<VPC_ID> \
  --query "Subnets[].[SubnetId,CidrBlock,AvailabilityZone,Tags[?Key=='Name']|[0].Value]" --output table

# Which route table has an Internet Gateway route, and which subnets use it?
aws ec2 describe-route-tables --filters Name=vpc-id,Values=<VPC_ID> \
  --query "RouteTables[].[RouteTableId,Associations[].SubnetId,Routes[].GatewayId]"
```

Broken if: the ALB's subnet IDs are the 10.0.1.0/24 and 10.0.3.0/24 ones, which are not associated with the route table that has igw-...

3. Issue 2: Health check path

```
aws elbv2 describe-target-groups \
  --query "TargetGroups[].[TargetGroupName,TargetGroupArn,HealthCheckPath,HealthCheckPort,HealthCheckProtocol]"
```

Broken if: the path shows /healthcheck.hmtl. It should be /healthcheck.html.

4. Issue 3: Web security group rules

```
aws ec2 describe-security-groups --filters Name=vpc-id,Values=<VPC_ID> \
  --query "SecurityGroups[].[GroupName,GroupId,IpPermissions[].[FromPort,ToPort,UserIdGroupPairs[].GroupId]]"
```

Broken if: websg only has port 22 with the ALB SG as the source, and there is no port 80 rule.

5. Issue 4: Is the ASG attached to the target group?

```
# ASG details: TargetGroupARNs should not be empty
aws autoscaling describe-auto-scaling-groups \
  --query "AutoScalingGroups[].[AutoScalingGroupName,TargetGroupARNs,Instances[].InstanceId]"
```

```
# Another way to check the attachment
aws autoscaling describe-load-balancer-target-groups --auto-scaling-group-name <ASG_NAME>

# Are any targets registered, and are they healthy?
aws elbv2 describe-target-health --target-group-arn <TARGET_GROUP_ARN>
```

Broken if: TargetGroupARNs is [], and describe-target-health returns no targets.

Bonus: check the listener

```
aws elbv2 describe-listeners --load-balancer-arn <ALB_ARN> \
  --query "Listeners[].[Port,Protocol,DefaultActions[].TargetGroupArn]"
```

This should show HTTP 80 forwarding to the stack's target group. It is fine in this lab, but worth confirming.

Order to check

Check in this order, which follows the traffic path from the internet to the instances: **Issue 1** (ALB subnets), then **Issue 4** (ASG attachment), then **Issue 2** (health check path), then **Issue 3** (security group). Once all four checks show a problem, start the fixes from the resolution guide.